

Build with Confidence: Ensure Quality and Throughput with AMVero using AI

Reduce Scrap, Increase Throughput, and Empower Your Team with a Smarter, Universal Build Process.

In additive manufacturing, monitoring a diverse fleet of printers can be complex.

AMVero changes that. It's a practical, straightforward monitoring tool that acts as an expert assistant for your entire team, giving you the power to prevent catastrophic errors, simplify quality control, and get part quality assurance from every machine, regardless of the manufacturer and the type of camera that is used.

AMVero's intelligent monitoring empowers your operators by handling the routine observation of thousands of flawless layers, allowing them to focus their expertise on higher-impact work like process optimization and quality assurance.

The AMVero Advantage: Intelligent, Simple, and Universal

We built AMVero to be different. Instead of locking you into a single ecosystem or requiring expensive, proprietary hardware, we provide a flexible solution that works for you.

AMVero is a minimally invasive solution meaning that machines do not need to be requalified / validated.

- **No Expensive Hardware Needed:** Get started without a major capital investment. AMVero leverages the in-situ monitoring cameras already widely available in today's industrial printers. There are no costly, proprietary sensors to install. Any third party hardware / modification can lead to significant downtime – don't take our word for it, leading OEM's can confirm this.
- **Intuitive From Day One:** We believe powerful software doesn't have to be complicated. AMVero features a clean, simple, and intuitive user interface designed for a quick learning curve, empowering your operators to become experts in hours, not weeks.
- **Truly Machine-Agnostic:** AMVero is designed for universal compatibility. It supports all LPBF (Laser Powder Bed Fusion) 3D printers out of the box, freeing you from OEM lock-in. Manage your entire multi-brand fleet from a single, unified platform.

How AMVero Solves Your Daily Challenges

- **Detect and Locate Errors in Real-Time:** Our AI-powered system doesn't just find errors, it locates them. It instantly detects process-killing issues like

warping, spatter, recoater hopping, short feed and recoater lines in the context of the as-built part. Your operator gets a simple and actionable alert for anomalies, showing the exact location on the digital model to help them quickly evaluate its severity.

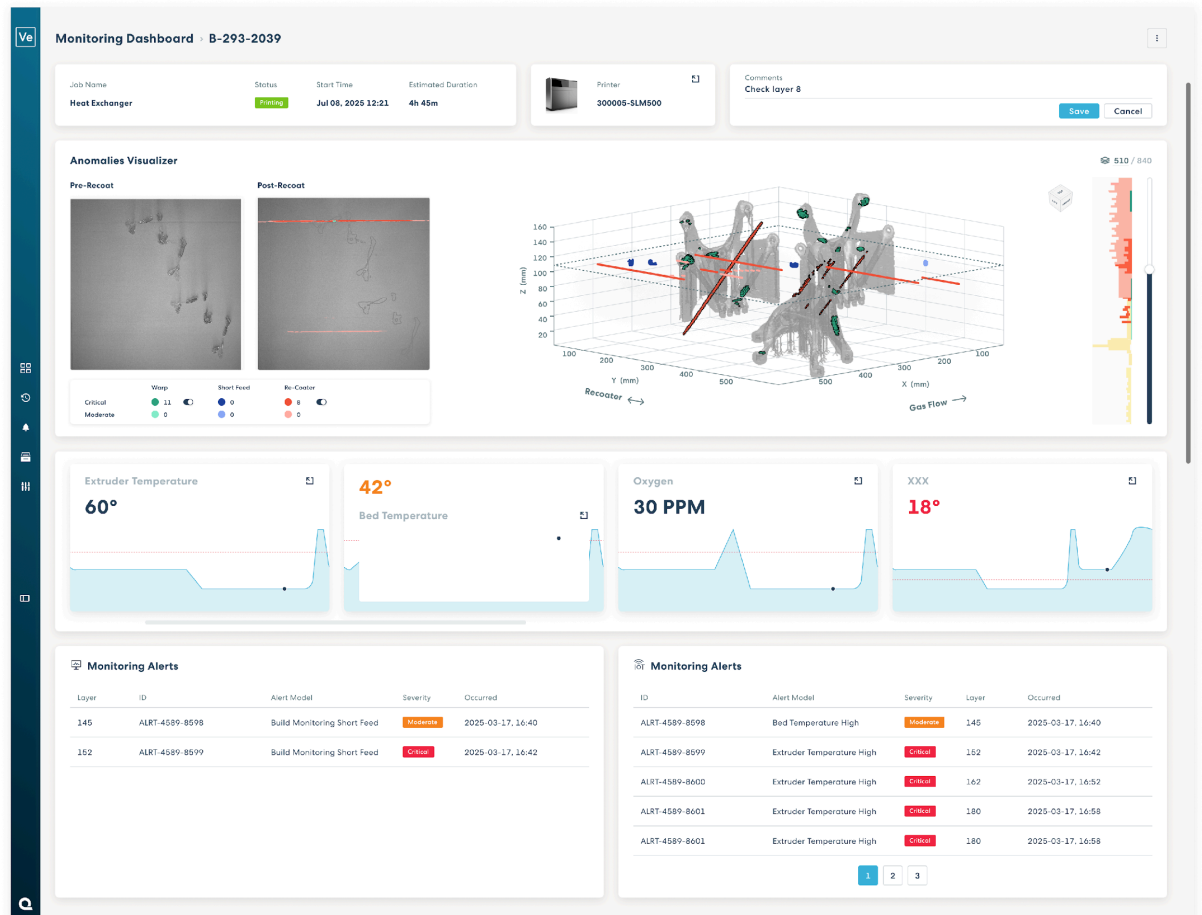


Figure 1 Ongoing build monitoring

- **Manage Your Entire Fleet from a Single Dashboard:** No more wasting time hunting for information. AMVero connects all your printers' critical manufacturing data into one clear, easy-to-use dashboard. Get a high-level overview of your entire fleet on a single screen and instantly identify which machines require attention. Create alert definitions, receive notifications, and generate configurable reports at the end of each build.

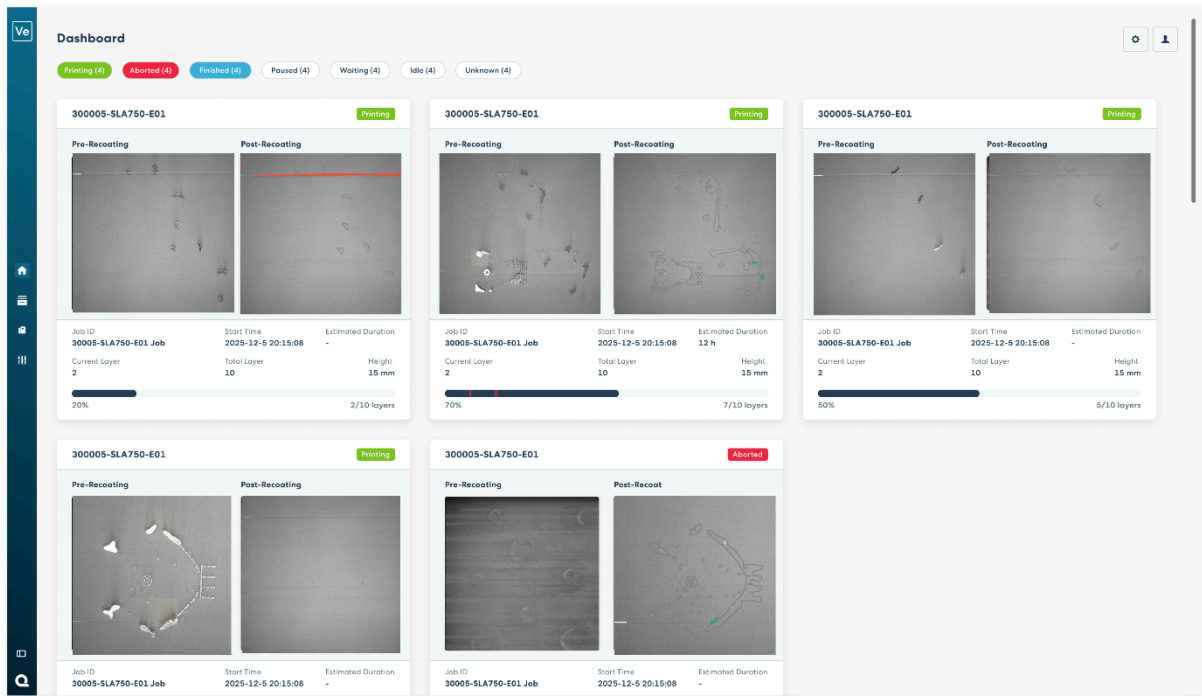


Figure 2 AMVero Real Time Dashboard

- Ensure Full Traceability:** Gain a complete, transparent record of the entire build process. AMVero automatically logs all sensor data, imagery, and detected anomalies, creating a comprehensive digital record for each part. This ensures end-to-end traceability, critical for quality assurance and certification.
- Streamline Root-Cause Analysis:** Use historical build data for in-depth failure analysis. When a build fails, you can easily export the detailed information to tools like 3DXpert Build Inspection, allowing your engineers to investigate the root cause and prevent future occurrences.
- Generate Configurable, Actionable Reports:** At the end of every job, AMVero automatically generates a comprehensive summary report. Configure these reports to fit your specific quality standards, highlighting the most severe anomalies for each part. This significantly reduces manual review time and makes post-build analysis faster and more intelligent.

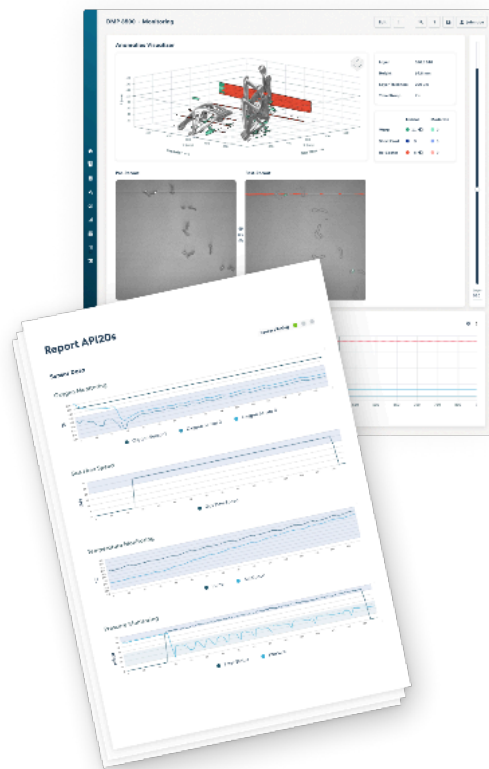


Figure 3 Customized end-of-build report

Ensure Repeatable Quality with Build Compare

Achieve unparalleled build-to-build consistency with the Build Compare tool. This powerful feature enhances process control by enabling a direct, data-driven comparison between a current build and an approved golden build. The primary benefit is the ability to instantly assess if a new build conforms to your established quality standards. By automatically isolating and quantifying only the significant deviations, AMVero streamlines the pass/fail decision-making process. This eliminates tedious manual comparisons and empowers you to approve parts with speed and confidence.

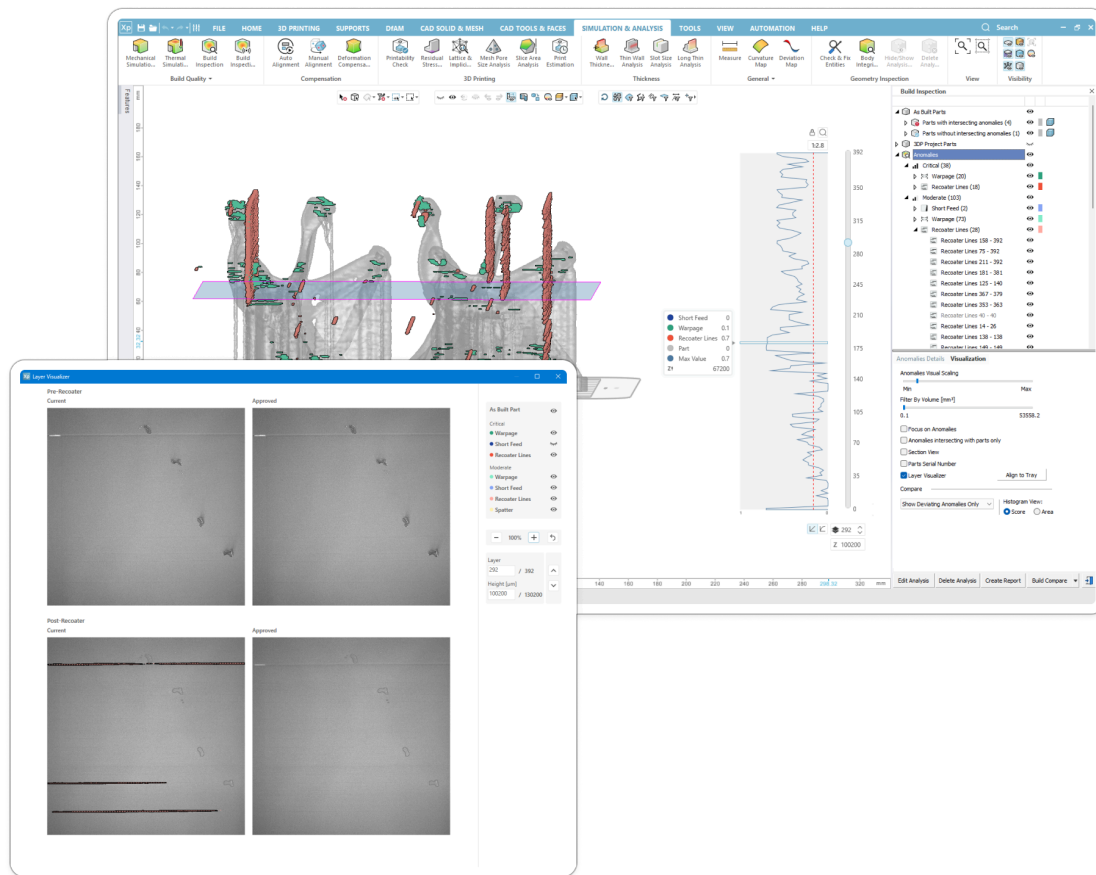


Figure 4 Build comparison

Trusted in Production by Industry Leaders

AMVero isn't just a concept; it's a production-proven solution. We are proud to partner with global energy technology leaders like Baker Hughes, who rely on AMVero in their demanding manufacturing environments to ensure part quality, reliability, and operational excellence.

Direct Benefits for Your Shop Floor

- Drastically Cut Scrap & Rework Costs:** By catching critical errors as they occur, you stop wasting expensive powder and machine hours on parts that are destined to fail.
- Save on Costly Post-Processing:** With enhanced in-process quality control and detailed part reports, you can reduce the need for expensive and time-consuming post-build inspections like CT scanning, saving both time and money.
- Increase Machine Uptime & Throughput:** Fewer failed builds and faster quality checks mean your machines are freed up for the next job sooner. Get more parts out the door without adding more equipment.

- **Empower Your Operators:** AMVero acts as a second pair of expert eyes for every operator. Simple, contextual alerts and clear reports boost their confidence and reduce the chance of human error.

Build Monitoring Benefits

Developed By **OQTON**

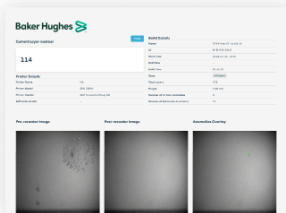
Baker Hughes

MONITORING BENEFIT

Saved

136 hours

per year per printer

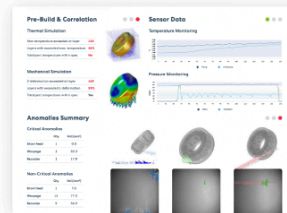


REPORT BENEFIT

Saved

>1 hour

per suspicious detection

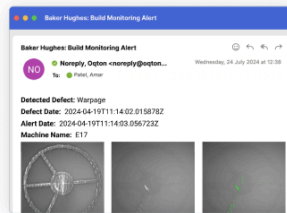


ALERT BENEFIT

Reduction of

-18 %

in scrap-related costs



See How It Works on Your Floor.

Stop letting build failures and analyzing scrap disrupt your schedule and budget. It's time to give your team the universal, production-proven AI based solution to improve your production throughput.

Learn more about our offering [here](#).